Referee package – DR-2 for the lattice class: an unconditional, decoupling-free additive-energy bound on the sphere

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This referee document concerns DR-2 (the additive-energy / discrete-restriction bound) for the lattice competitor class. It is a harmonic-analysis / additive-combinatorics result, of interest independently of TECT. **Scope:** $Q = \Lambda \cap \{|x|^2 = R\}$ a fixed crystallographic lattice shell. **Not claimed:** the arbitrary (non-lattice) Q case remains OPEN (the decoupling-based route is T4 conditional and not load-bearing); this package does not by itself close the full Reading-H C_{full} theorem. **Grade:** T7 CONFIRMED on the lattice class (operator promotion 2026-06-12), with the standard textbook NT import accepted as an annotation, not a conditionality: the clean-run is operator-confirmed (8/8 scripts, 38/38 asserts) and the Lemma NT pin is in-bundle (Sec. 3: proof note `dr2-lemma-nt-inbundle-260612-v1.0` – unramified case proved in full, ramified corner standard-cited + outlined + exhaustively machine-covered on the applied domain).

2. Definitions and the result

Definitions (patch v1.1-2). Let $Q \subset \{|x|^2 = R\}$ be finite with $N = |Q|$. The additive energy and the sum-representation counts are

$$E_+(Q) = \#\{(a,b,c,d) \in Q^4 :\ a+b=c+d\}, \qquad r_Q(m) = \#\{(a,b) \in Q^2 :\ a+b=m\},$$

and the maximal proper sum-circle occupancy is $T'(Q) = \max_{m \neq 0} r_Q(m)$. Lemma A (sum-circle richness) is the chain

$$E_+(Q) = \sum_m r_Q(m)^2 \ \le\ r_Q(0)^2 + T'(Q) \sum_{m \neq 0} r_Q(m) \ \le\ (1 + T'(Q))N^2,$$

using $r_Q(0) \le N$ and $\sum_m r_Q(m) = N^2$.

The result. For $Q = \Lambda \cap \{|x|^2 = R\}$ (Λ a fixed crystallographic lattice, rational after clearing denominators), the additive energy obeys

$$E_+(Q) \ \le\ (1 + C_\Lambda \max_{m \neq 0} d(4R - |m|^2))N^2 \ \le\ (1 + C_\Lambda C_\varepsilon R^\varepsilon)N^2,$$

decoupling-free, where $d(\cdot)$ is the divisor function and C_Λ is the representation constant of Lemma NT below ($C_\Lambda \le 6$ once the textbook lemma is imported). For Gauss-typical shells $R \sim N^2$ this gives, after redefining ε ($R^\varepsilon \sim N^{2\varepsilon}$), $E_+ \le_\varepsilon N^{2+\varepsilon}$.

3. Structure of the argument

Circle-to-binary-form reduction (patch v1.1-3). Fix $m \neq 0$ and suppose $a + b = m$ with $|a|^2 = |b|^2 = R$. From $b = m - a$, $R = |b|^2 = |m|^2 - 2a \cdot m + |a|^2$ forces $2a \cdot m = |m|^2$. Substituting $y = 2a - m$,

$$y \cdot m = 2a \cdot m - |m|^2 = 0, \qquad |y|^2 = 4|a|^2 - 2\,(2a \cdot m) + |m|^2 = 4R - |m|^2, \qquad y \equiv -m \pmod 2,$$

and $a \mapsto y$ is injective. Hence $r_Q(m)$ is bounded by the HOMOGENEOUS representation count of $D = 4R - |m|^2$ on the rank-two lattice section $L_m = \{y \in \Lambda_{\text{int}} : \ y \cdot m = 0\}$ (machine check: `dr2_divcirc_proof.py`, substitution exact, injectivity verified).

Lemma NT (textbook import; patch v1.1-4). *For every rank-two lattice section L_m as above, with induced positive-definite integral binary quadratic form q of content g and primitive part q_0 of discriminant $\Delta_0 < 0$,*

$$r_{L_m}(D) = \#\{y \in L_m : \ |y|^2 = D\} \ \le \ C_{\text{NT}}\, d(D), \qquad C_{\text{NT}} = w(\Delta_0) \le 6,$$

*where $w(\Delta_0)$ is the number of automorphs (*6 *for $\Delta_0 = -3$,* 4 *for $\Delta_0 = -4$,* 2 *otherwise).* Provenance: $r_q(D) = r_{q_0}(D/g)$ (zero unless $g \mid D$), and Dirichlet's representation formula gives the TOTAL count over the full form class group of discriminant Δ_0 as $w(\Delta_0) \sum_{d|D'} \chi_{\Delta_0}(d) \le w(\Delta_0)\, d(D') \le w(\Delta_0)\, d(D)$; a single class's count is at most the total. This is textbook material (Dirichlet's class-number/representation formula; e.g. Landau, or Cox, *Primes of the Form $x^2 + ny^2$*), but the package treats it as an IMPORT: the load-bearing chain is written with the constant C_Λ, and $C_\Lambda \le 6$ holds once Lemma NT is cited or proved in the bundle. **Pin status (v1.4):** the lemma is now PINNED IN-BUNDLE by the proof note `dr2-lemma-nt-inbundle-260612-v1.0`: the unramified case (every odd $p \mid D$ with $p^2 \nmid \Delta_0$) is proved in full, elementarily, with the explicit constant $r_{L_m}(D) \le 24\, d(D)^2$ (sufficient for the chain, which needs only $d(D)^2 \ll_\varepsilon D^\varepsilon$); the ramified corner carries a book-level citation (Landau/Cassels/Cox), a conductor-descent outline, and EXHAUSTIVE applied-domain machine coverage. Machine evidence: $r \le 6\,d$ holds for EVERY distinct sum m of every applied shell – 2,644,976 sums, worst ratio 0.250, 24,096 ramified instances all passing (worst 0.119) (`dr2_lemma_nt_exhaustive.py` 5/5); the richest-circle spot check (`dr2_divcirc_proof.py` 4/4) is thereby subsumed.

Assembly. Lemma A + the reduction + Lemma NT give $T'(Q) \le C_\Lambda d(4R - |m^*|^2) \ll_\varepsilon R^\varepsilon$ (divisor bound, uniform in m), hence the stated result — no ℓ^2-decoupling is used. The conditional decoupling route (`dr2_decoupling_iteration.py` / `dr2_decoupling_exponent.py`) is retained as a CROSS-CHECK only and is not load-bearing.

4. Numerical reproduction

Entry-script manifest (patch v1.1-1; amended v1.4: 8 scripts). Eight scripts, each self-contained with self-test asserts, all exit 0 (operator clean-run CONFIRMED 2026-06-12: 8/8 scripts, 38/38 asserts PASS; the 8th, `dr2_lemma_nt_exhaustive.py`, was added with the v1.4 Lemma NT pin, 1.2 s):

1. `codes/vacuum/dr2_circle_richness.py` (5/5) — Lemma A (sum-circle richness) and Corollary 1.

2. `codes/vacuum/dr2_divcirc_proof.py` (4/4) — the substitution $y = 2a - m$ (exact, injective), $T' \le r_{L_m}(4R - |m|^2) \le 6\,d(4R - |m|^2)$, worst $T'/6d = 0.25$.

3. `codes/vacuum/dr2_lattice_divisor.py` (7/7; v2.0.0 runtime patch) — lattice-divisor scaling: T'/N falls $0.107 \to 0.024$ across shells $R = 101$–9974; $E_+/N^2 \le 5.3$. The v1.0.0 occupancy loop timed out on $R = 9974$ in referee clean-runs; v2.0.0 replaces it by one vectorized unique-count pass via the exact full-shell identity $\#(Q \cap C_m) = r_Q(m)$, machine-verified against the retained

v1.0.0 reference on $R \le 909$ and pinned to the archived $R = 9974$ values ($N = 2040$, $T' = 48$, $E_+ = 16291944$) as a regression oracle. Runtime 2.5 s.

4. `codes/vacuum/dr2_cross_energy_lemma.py` (6/6) — the bilinear cross-energy lemma $E_+(A, B) \le 3|A||B|$ and the off-diagonal pencil bound.
5. `codes/vacuum/dr2_weighted_energy.py` (3/3) — the weighted Lemma A ($\sum_t w_t^2 \le (1+T')\|c\|_2^4$).
6. `codes/vacuum/dr2_decoupling_iteration.py` (4/4) — decoupling-route cross-check (NOT load-bearing).
7. `codes/vacuum/dr2_decoupling_exponent.py` (4/4) — decoupling-route exponent cross-check (NOT load-bearing).
8. `codes/vacuum/dr2_lemma_nt_exhaustive.py` (5/5; NEW v1.4) — the Lemma NT applied chain $r_Q(m) \le 6\,d(4R - |m|^2)$ verified EXHAUSTIVELY for all 2,644,976 distinct sums across the six shells, ramified corner included; 1.2 s.

Supporting / integration-layer (NOT an entry script): `codes/vacuum/dr2_hadmcoh_margin.py` — the H-ADM-COH discharge-decision margin (long-running; belongs to the `dr2-hadmcoh-discharge-decision` integration note, run artefact `runs/260608-dr2-hadmcoh-margin/`); not load-bearing for the lattice-class theorem. Manifest correction note: the v1.0 "7 scripts" was an auto-discovered list that dropped `dr2_lattice_divisor.py` (a single-line regex artefact: the headline note lists it on the second line of its reproduction command) and included `dr2_hadmcoh_margin.py`; the present manifest is the corrected, load-bearing-plus-declared-cross-check set. Run the entry manifest from the bundle and diff against `expected/`.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it. In particular: lattice shells only; arbitrary-Q OPEN; decoupling route T4 conditional cross-check; no full C_{full} claim.

(β) *Attack the tier / over-claim.* The sharpest objection is the $6\,d(D)$ representation bound: "Dirichlet class-number formula" alone is not a citation. ADDRESSED in v1.1 (operator-directed): Lemma NT is stated precisely with its provenance, the load-bearing chain uses the relaxed constant C_Λ, and the T7 label is precise (footer): T7 CONFIRMED on lattice shells with the STANDARD NT import accepted (operator ruling 2026-06-12) – the unramified case is in-bundle proved and the ramified corner is standard-cited + exhaustively machine-covered. The claim-ledger tier is the arbiter.

(γ) *Attack reproduction / self-containment.* Each entry script is self-contained (no runtime run-artefact reads — audited after the SC-SCOPE joint_pairing lesson), carries self-test asserts, and ships in the bundle with its transitive dependencies; a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the Lemma-A chain is re-derivable from its definitions ($\sum_m r_Q(m) = N^2$ exactly; $r_Q(0) \le N$); the substitution identity $|2a - m|^2 = 4R - |m|^2$ is checked exactly on every tested configuration; the worst machine ratio $T'/6d = 0.25$ leaves a $\times 4$ headroom below the Lemma NT ceiling; and the falsifier (footer) is concrete and pre-registered (a lattice shell violating the divisor bound, or a counterexample to the rank-2 representation count).

6. Result footer

```
Result ID:               B5-BEYOND-LAYER-BOUND / DR-2 lattice-class
                         referee package
Precise statement:       For Q=Lambda cap {|x|^2=R} (fixed
                         crystallographic lattice), with N=|Q|,
                         E_+(Q)=#{a+b=c+d}, r_Q(m)=#{(a,b):a+b=m},
                         T'(Q)=max_{m!=0} r_Q(m):
                         E_+(Q) <= (1+T'(Q))N^2 (Lemma A),
                         T'(Q) <= C_Lambda max_{m!=0} d(4R-|m|^2)
                         (y=2a-m reduction + Lemma NT; C_Lambda<=6
                         once the textbook lemma is imported),
                         hence E_+(Q) <=_eps R^eps N^2, decoupling-
                         free; Gauss-typical R~N^2 gives E_+ <=_eps
                         N^{2+eps} (eps redefined, R^eps~N^{2eps}).
Scope:                   lattice shells Q=Lambda cap {|x|^2=R} ONLY.
                         Arbitrary non-lattice Q OPEN; decoupling
                         route T4 conditional (cross-check, not
                         load-bearing); not a full C_full closure.
Evidence grade:          T7 CONFIRMED on lattice shells (operator
                         promotion 2026-06-12); assumption class:
                         standard textbook NT import accepted
                         (annotation, not conditionality). Lemma NT
                         pinned in-bundle (unramified: full proof,
                         24 d(D)^2; ramified corner: standard-cited
                         + outlined + exhaustively machine-covered).
                         EXECUTED (8 entry scripts, 38/38 asserts;
                         2,644,976 sums, worst 0.250).
Reproduction command:    python codes/vacuum/dr2_circle_richness.py ;
                         codes/vacuum/dr2_divcirc_proof.py ;
                         codes/vacuum/dr2_lattice_divisor.py ;
                         codes/vacuum/dr2_cross_energy_lemma.py ;
                         codes/vacuum/dr2_weighted_energy.py ;
                         codes/vacuum/dr2_decoupling_iteration.py ;
                         codes/vacuum/dr2_decoupling_exponent.py ;
                         codes/vacuum/dr2_lemma_nt_exhaustive.py
Falsifier:               a lattice shell with E_+ exceeding the
                         divisor bound; or a counterexample to the
                         rank-2 representation count r <= C_NT d(D).
Tier before / after:     DR2-Lattice-T7Candidate-260612 ->
                         DR2-Lattice-T7-NTstandard-260612 (operator
                         promotion 2026-06-12): T7 CONFIRMED on
                         lattice shells, PUBLISHED-BUNDLE CONFIRMED;
                         8/8 scripts, 38/38 asserts. No B5
                         claim-tier promotion by this package alone.
No-overclaim statement:  lattice class only, modulo the explicitly
                         stated Lemma NT (constant relaxed to
                         C_Lambda in the load-bearing chain);
                         arbitrary non-lattice Q is OPEN; the
                         decoupling route is a non-load-bearing
                         cross-check.
Next required action:    none for this package (T7 CONFIRMED,
                         PUBLISHED). Mainline per policy sec.15:
                         T-013 claim-level SYNTHESIS capstone.
                         Separately tracked: arbitrary non-lattice
                         Q (OPEN); full STEP-5B budget comparison
                         + admissibility gates.
```